Curved Yang-Mills gauge theories

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Abstract*

We construct a gauge theory based on principal bundles $\mathscr{P}$ equipped with a right $\mathscr{G}$-action, where $\mathscr{G}$ is a Lie group bundle instead of a Lie group. Due to the fact that a $\mathscr{G}$-action acts fibre by fibre, pushforwards of tangent vectors via a right-translation act now only on the vertical structure of $\mathscr{P}$. Thus, we generalize pushforwards using a connection on $\mathscr{G}$ which will modify the pushforward. A horizontal distribution on $\mathscr{P}$ invariant under such a modified pushforward will provide a proper notion of Ehresmann connection. For achieving gauge invariance we impose conditions on the connection 1-form μ on $\mathscr{G}$: μ has to be a multiplicative form, *i.e.* closed w.r.t. a certain simplicial differential δ on $\mathscr{G}$, and the curvature R_μ of μ has to be δ-exact with primitive ζ; μ will be the generalization of the Maurer-Cartan form of the classical gauge theory, while the δ-exactness of R_μ will generalize the role of the Maurer-Cartan equation. This introduces the notion of multiplicative Yang-Mills connections, a connection which helped classifying singular foliations and symmetry breaking. For allowing curved connections on $\mathscr{G}$ in the dynamical theory we will need to generalize the typical definition of the curvature/field strength F on $\mathscr{P}$ by adding ζ to F.

Several examples for a gauge theory with a curved μ will be provided, including the inner group bundle of the Hopf fibration $\mathbb{S}^7 \to \mathbb{S}^4$, and a classification for gauge theories with structural simple group bundles will be provided, including a classification for whether these theories admit a classical description.

Last but not least, this paper aims to accessible for beginners which is why it only assumes knowledge of classical gauge theory.

*Abbreviations used in this paper: **LGB(s)** for Lie group bundle(s), **LAB(s)** for Lie algebra bundle(s).

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1. Introduction and summary

2. Lie group bundles with connection

We will construct a gauge theory on a principal bundle equipped with an LGB-action. Thus let us first study LGBs; since we will also generalise the Maurer-Cartan form, we will study connections on LGBs with a suitable curvature equation.

That is, we will consider LGBs $\pi_{\mathscr{G}}\colon \mathscr{G} \to M$ over a smooth manifold M and we will denote LGB actions on a manifold N along a map $\Phi\colon N \to M$ as in the following:

Definition 2.1: Lie group bundle actions

for example [1, §1.6, special case of Def. 1.6.1, page 34]

Let M, N be smooth manifolds, $\pi_{\mathscr{G}}\colon \mathscr{G} \to M$ an LGB over M, and $\Phi\colon N \to M$ a smooth map. Then a **right-action of $\mathscr{G}$ on N** is a smooth map

$$\begin{aligned} N \,{}_{\Phi}\!\times_{\pi_{\mathscr{G}}} \mathscr{G} &\to N, \\ (p, g) &\mapsto p \cdot g, \end{aligned}$$

satisfying the following properties:

$$\begin{aligned} \Phi(p \cdot g) &= \pi(g), \\ (p \cdot g) \cdot h &= p \cdot (gh), \\ p \cdot \mathbb{1}_{\Phi(p)} &= p \end{aligned}$$

for all $p \in N$ and $g, h \in \mathscr{G}_{\Phi(p)}$, where $\mathbb{1}$ is the neutral section of $\mathscr{G}$. Similarly one defines left actions.

In this work we will need to apply the tangent map to an LGB action, and for doing so the following proposition will be useful; for this remember that it is a standard exercise to show that the tangent bundle of the pullback $N \,{}_{\Phi}\!\times_{\pi_{\mathscr{G}}} \mathscr{G}$ is given by just applying the tangent functor T on all components, that is,

$$\mathrm{T}\big(N \,{}_{\Phi}\!\times_{\pi_{\mathscr{G}}} \mathscr{G}\big) = \mathrm{T}N \,{}_{\mathrm{T}\Phi}\!\times_{\mathrm{T}\pi_{\mathscr{G}}} \mathrm{T}\mathscr{G} \ ,$$

where the right side is naturally viewed as a bundle over $N \,{}_{\Phi}\!\times_{\pi_{\mathscr{G}}} \mathscr{G}$.

Proposition 2.2: Differential of smooth LGB actions

Let M and N be smooth manifolds, $\pi_{\mathscr{G}}\colon \mathscr{G} \to M$ an LGB over M, $\Phi\colon N \to M$ a smooth map, and assume we have a right $\mathscr{G}$-action on N, $\Psi\colon N \,{}_{\Phi}\!\times_{\pi_{\mathscr{G}}} \mathscr{G} \to N$. Then we have

$$\mathrm{T}_{(p,g)}\Psi(X, Y) = \mathrm{T}_p r_\sigma(X) + \overline{(\mu_{\mathscr{G}})_g\big(Y - \mathrm{T}_x\sigma(\omega)\big)}\Big|_{p\cdot g}$$

for all $(p, g) \in N \,{}_{\Phi}\!\times_{\pi_{\mathscr{G}}} \mathscr{G}$ and $(X, Y) \in \mathrm{T}N \,{}_{\mathrm{T}\Phi}\!\times_{\mathrm{T}\pi_{\mathscr{G}}} \mathrm{T}\mathscr{G}$, where $x := \Phi(p) = \pi_{\mathscr{G}}(g)$, σ is any (local) section of $\mathscr{G}$ with $\sigma_x = g$, and ω is an element of $\mathrm{T}_x M$ given by

$$\omega := \mathrm{T}_p\Phi(X) = \mathrm{T}_g\pi_{\mathscr{G}}(Y) \ .$$

Proof. Due to $N * \mathcal{G} = f^*\mathcal{G}$ we are going to use Lemma **??**. That is, fix $(p, g) \in N * \mathcal{G}$ and $X \in \mathrm{T}_p N$, $Y \in \mathrm{T}_g \mathcal{G}$ with

$$\mathrm{D}_p f(X) = \mathrm{D}_g \pi(Y) =: \omega \in \mathrm{T}_{f(p)} M.$$

Recall that we can localize LGB actions in sense of Rem. **??**; so, let $x := f(p) = \pi(g)$, and fix a trivialization of $\mathcal{G}$ around x. Then it is clear that there is a (local)[2] section $\sigma \in \Gamma(\mathcal{G})$ with $\sigma_x = g$. Observe that we can write

$$Y = Y - \mathrm{D}_x \sigma(\omega) + \mathrm{D}_x \sigma(\omega).$$

The two first summands result into a vertical tangent vector due to

$$\mathrm{D}_g \pi\big(Y - \mathrm{D}_x \sigma(\omega)\big) = \mathrm{D}_g \pi(Y) - \mathrm{D}_x \underbrace{(\pi \circ \sigma)}_{=\mathbb{1}_M}(\omega) = \mathrm{D}_g \pi(Y) - \omega = 0.$$

For the following recall the notations introduced in Def. **??** and **??**; we can derive that

$$\mathrm{D}_{e_x} R_\sigma \circ \mathrm{D}_x e = \mathrm{D}_x \underbrace{(R_\sigma \circ e)}_{x \mapsto e_x \sigma_x = \sigma_x} = \mathrm{D}_x \sigma.$$

So, we can also write

$$Y^v := Y - \mathrm{D}_x \sigma(\omega) = Y - \mathrm{D}_{e_x} R_\sigma \big(\mathrm{D}_x e(\omega)\big).$$

We have proven that $Y^v \in \mathrm{V}_g \mathcal{G}$, and thus $(0_p, Y^v) \in \mathrm{T}_{(p,g)}(f^*\mathcal{G})$ by Lemma **??**, where 0_p is the zero tangent vector of $\mathrm{T}_p N$. In the same fashion we also have

$$\big(X, \mathrm{D}_{e_x} R_\sigma \big(\mathrm{D}_x e(\omega)\big)\big) \in \mathrm{T}_{(p,g)}(f^*\mathcal{G})$$

because of

$$\mathrm{D}_g \pi\Big(\mathrm{D}_{e_x} R_\sigma \big(\mathrm{D}_x e(\omega)\big)\Big) = \mathrm{D}_g \pi(Y - Y^v) = \mathrm{D}_g \pi(Y) = \mathrm{D}_p f(X),$$

thence we can write in $\mathrm{T}_{(p,g)}(f^*\mathcal{G})$

$$(X, Y) = \big(X, \mathrm{D}_{e_x} R_\sigma \big(\mathrm{D}_x e(\omega)\big) + Y^v\big) = \big(X, \mathrm{D}_{e_x} R_\sigma \big(\mathrm{D}_x e(\omega)\big)\big) + (0_p, Y^v).$$

Hence also

$$\mathrm{D}_{(p,g)} \Phi(X, Y) = \mathrm{D}_{(p,g)} \Phi\big(X, \mathrm{D}_{e_x} R_\sigma \big(\mathrm{D}_x e(\omega)\big)\big) + \mathrm{D}_{(p,g)} \Phi(0_p, Y^v) \tag{1}$$

The second summand is quickly calculated as

$$\mathrm{D}_{(p,g)} \Phi(0_p, Y^v) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \Phi(p, \gamma) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \underbrace{(p \cdot \gamma)}_{\Phi_p(\gamma)} = \mathrm{D}_g \Phi_p(Y^v)$$

where $\gamma : I \to \mathcal{G}_x$ (I open interval of $\mathbb{R}$ containing 0) is a curve with $\gamma(0) = g$ and $\mathrm{d}/\mathrm{d}t|_{t=0} \gamma = Y^v$; this is due to the verticality of Y^v, and so $\mathrm{D}_g \Phi_p(Y^v)$ is also well-defined since Φ_p is a map $\mathcal{G}_x \to N$. Now recall Def. **??** and **??**, and then observe that we can write

$$\mathrm{D}_g \Phi_p(Y^v) = \big(\mathrm{D}_g \Phi_p \circ \mathrm{D}_{e_x} L_g \circ \mathrm{D}_g L_{g^{-1}}\big)(Y^v) = \mathrm{D}_{e_x} \underbrace{(\Phi_p \circ L_g)}_{\mathcal{G} \ni q \mapsto p \cdot g q = \Phi_{p \cdot g}(q)} \Big((\mu_{\mathcal{G}})_g (Y^v)\Big) = \overline{(\mu_{\mathcal{G}})_g (Y^v)}\Big|_{p \cdot g} \tag{2}$$

making use of the verticality of Y^v such that operators like $\mathrm{D}L_g$ can act on Y^v.

For the first summand in Eq. (1) we use

$$\begin{aligned}
\mathrm{D}_{(p,g)} \Phi\Big(X, \mathrm{D}_{e_x} R_\sigma \big(\mathrm{D}_x e(\omega)\big)\Big) &= \mathrm{D}_{(p,e_x)} \underbrace{\big(\Phi \circ (\mathbb{1}_N, R_\sigma)\big)}_{N * \mathcal{G} \ni (p,g) \mapsto p \cdot g \sigma_x = r_\sigma(p \cdot g)} \big(X, \mathrm{D}_x e(\omega)\big) \\
&= \mathrm{D}_{(p,e_x)} (r_\sigma \circ \Phi) \big(X, \mathrm{D}_x e(\omega)\big) \\
&= \mathrm{D}_p r_\sigma \Big(\mathrm{D}_{(p,e_x)} \Phi\big(X, \mathrm{D}_x e(\omega)\big)\Big)
\end{aligned}$$

[2]For simplicity of notation we omit the notation of restricting on some open subset of M.

$$= \mathrm{D}_p r_\sigma\Big(\mathrm{D}_{(p,e_x)}\Phi\big(X, \mathrm{D}_p(e\circ f)(X)\big)\Big)$$
$$= \mathrm{D}_p r_\sigma\Big(\mathrm{D}_{(p,p)}\big(\Phi\circ(\mathbb{1}_N, e\circ f)\big)(X,X)\Big),$$

but on the diagonal $\mathrm{diag}(N\times N)$ of $N\times N$ we have

$$\Big[\mathrm{diag}(N\times N)\ni(p,p)\mapsto\big(\Phi\circ(\mathbb{1}_N, e\circ f)\big)(p,p) = \Phi\big(p, e_{f(p)}\big) = p\cdot e_{f(p)} = p\Big] = \mathrm{pr}_i|_{\mathrm{diag}(N\times N)},$$

where pr_i is just the projection onto any of both i-th components ($i\in\{1,2\}$) in $N\times N$. Due to $(X,X)\in\mathrm{T}_{(p,p)}\big(\mathrm{diag}(N\times N)\big)$ we derive

$$\mathrm{D}_{(p,p)}\big(\Phi\circ(\mathbb{1}_N, e\circ f)\big)(X,X) = \mathrm{D}_{(p,p)}\mathrm{pr}_i(X,X) = X.$$

So, we get in total

$$\mathrm{D}_{(p,g)}\Phi\Big(X, \mathrm{D}_{e_x}R_\sigma\big(\mathrm{D}_x e(\omega)\big)\Big) = \mathrm{D}_p r_\sigma(X),$$

therefore

$$\mathrm{D}_{(p,g)}\Phi(X,Y) = \mathrm{D}_p r_\sigma(X) + \overline{(\mu_{\mathcal{G}})_g(Y^v)}\Big|_{p\cdot g}.$$

■

2.1. **Ehresmann connections ...**

2.1.1. *... as parallel transport.* In the following we will initially define connections via their corresponding parallel transport, for which we will assume that they are complete. That is, for a curve $\gamma\colon I\to M$ (I an open interval of $\mathbb{R}$ containing 0) we denote a parallel transport on fibre bundles $\pi_{\mathscr{F}}\colon\mathscr{F}\to M$ by

$$\mathrm{PT}^{\mathscr{F}}_\gamma\colon\mathscr{F}_{\gamma(0)}\to\Gamma(\gamma^*\mathscr{F}),$$
$$p\mapsto\mathrm{PT}^{\mathscr{F}}_\gamma(p),$$

similarly for the parallel transport on $\mathscr{G}$. We expect a sort of equivariance for the parallel transport of $p\cdot g$:

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List of References

[1] Kirill C. H. Mackenzie. General theory of Lie groupoids and Lie algebroids. *London Mathematical Society Lecture Note Series*, 213:xxxviii+501, 2005. https://doi.org/10.1017/CBO9781107325883.